



COMPLETE FUN GUIDE TO PROBABILITY DISTRIBUTIONS (CO2) ✨



DISCRETE DISTRIBUTIONS

Used when your random variable takes on finite or countable values (like counting things).



1. Discrete Uniform Distribution

All outcomes are equally likely to happen.

- **PMF:** $f(x) = \frac{1}{n}$
- **Mean:** $\mu = \frac{n+1}{2}$
- **Variance:** $\text{Var}(X) = \frac{n^2-1}{12}$



2. Binomial Distribution (Super Important!)

Used when: You have a fixed number of trials, only two possible outcomes (Success/Fail, Head/Tail), and the trials are independent.

- **Binomial PMF:** $P(X=x) = \binom{n}{x} p^x q^{n-x}$
 - (Note: $q = 1 - p$)
- **Mean:** $E(X) = np$
- **Variance:** $\text{Var}(X) = npq$



3. Poisson Distribution

Used for: Rare events occurring over a fixed interval of time or space. **Examples:** Accidents, phone calls, manufacturing defects. **Poisson Process Conditions:** Events are independent, have a constant average rate, and occur randomly.

- **PMF:** $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$
- **Mean & Variance:** They are magical because they are both equal!
 - $E(X) = \text{Var}(X) = \lambda$



CONTINUOUS DISTRIBUTIONS

Used when your random variable can take any value within a continuous interval (like measuring height, weight, or time).



1. Continuous Uniform Distribution

- **PDF:** $f(x) = \frac{1}{b-a}$ for $a < x < b$
- **Mean:** $E(X) = \frac{a+b}{2}$
- **Variance:** $\text{Var}(X) = \frac{(b-a)^2}{12}$



2. Exponential Distribution

Used for: Modeling waiting times or duration between events.

- **PDF:** $f(x) = \lambda e^{-\lambda x}$
- **CDF:** $F(x) = 1 - e^{-\lambda x}$
- **Mean:** $E(X) = \frac{1}{\lambda}$
- **Variance:** $Var(X) = \frac{1}{\lambda^2}$

🔔 3. Normal (Gaussian) Distribution (The MVP)

The most important continuous distribution, easily recognized by its perfectly symmetrical Bell-shaped curve



- **PDF:** $f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
- **Standard Normal Variable (\$Z\$):** Used to convert any normal distribution to a standard one so you can use Z-tables!
 - $Z = \frac{X-\mu}{\sigma}$
- **The Empirical Rule (Approximate percentages):**
 - ✅ 68% of data falls within 1σ of the mean.
 - ✅ 95% of data falls within 2σ of the mean.
 - ✅ 99.7% of data falls within 3σ of the mean.

🧠 APPROXIMATIONS & POWERFUL THEOREMS

🔄 Normal Approximation to Binomial

If you are doing a Binomial problem but n is very large and p is not too small, you can swap to a Normal distribution:

- $B(n, p) \approx N(np, npq)$

☀️ Central Limit Theorem (CLT)

One of the most powerful theorems in all of statistics!

- **It states:** The sum or average of many independent random variables will become approximately a Normal Distribution.
- **The Magic:** This happens even if the original distribution is not normal at all!

⚖️ Statistical Inequalities

When you don't know the exact distribution, these inequalities save the day!

- **Chebyshev Inequality:** $P(|X-\mu| < k\sigma) \geq 1 - \frac{1}{k^2}$
- **Markov Inequality (For non-negative variables):** $P(X \geq a) \leq \frac{E(X)}{a}$

📄 FULLY SOLVED CO2 NUMERICALS 🔥

1. Binomial Distribution: Expected Heads

- **Question:** 5 coins are tossed. Find the expected number of heads.
- **Data Given:**

- $n = 5$ (number of trials)
- $p = 1/2$ (probability of getting heads on a single coin)
- **Solution:**
 - We know the formula for the mean (expected value) of a Binomial distribution is $E(X) = np$.
 - $E(X) = 5 \times \frac{1}{2} = 2.5$
 - **Answer:** The expected number of heads is 2.5.

2. Poisson Distribution: Call Center Traffic

- **Question:** In a telephone exchange, the average number of calls per minute is $\lambda = 2.5$. Find the probability of getting exactly 4 calls in a given minute.
- **Data Given:**
 - $\lambda = 2.5$
 - $x = 4$
- **Solution:**
 - Using the Poisson PMF formula: $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$
 - Substitute the values: $P(X=4) = \frac{e^{-2.5} (2.5)^4}{4!}$
 - (Note: Calculate $e^{-2.5}$ and $(2.5)^4$ using your scientific calculator during the exam!)

3. Normal Distribution: Z-Score Conversion for Heights

- **Question:** The mean height of a group is 64.5 with a standard deviation of 4.5. Find the percentage of people falling between specific heights.
- **Data Given:**
 - $\mu = 64.5$
 - $\sigma = 4.5$
- **Solution Strategy:**
 - Convert your target height ' X ' into a Z-score using the formula: $Z = \frac{X - \mu}{\sigma}$
 - Once you calculate the Z value, use the Standard Normal Distribution Table to find the corresponding probabilities/percentages.